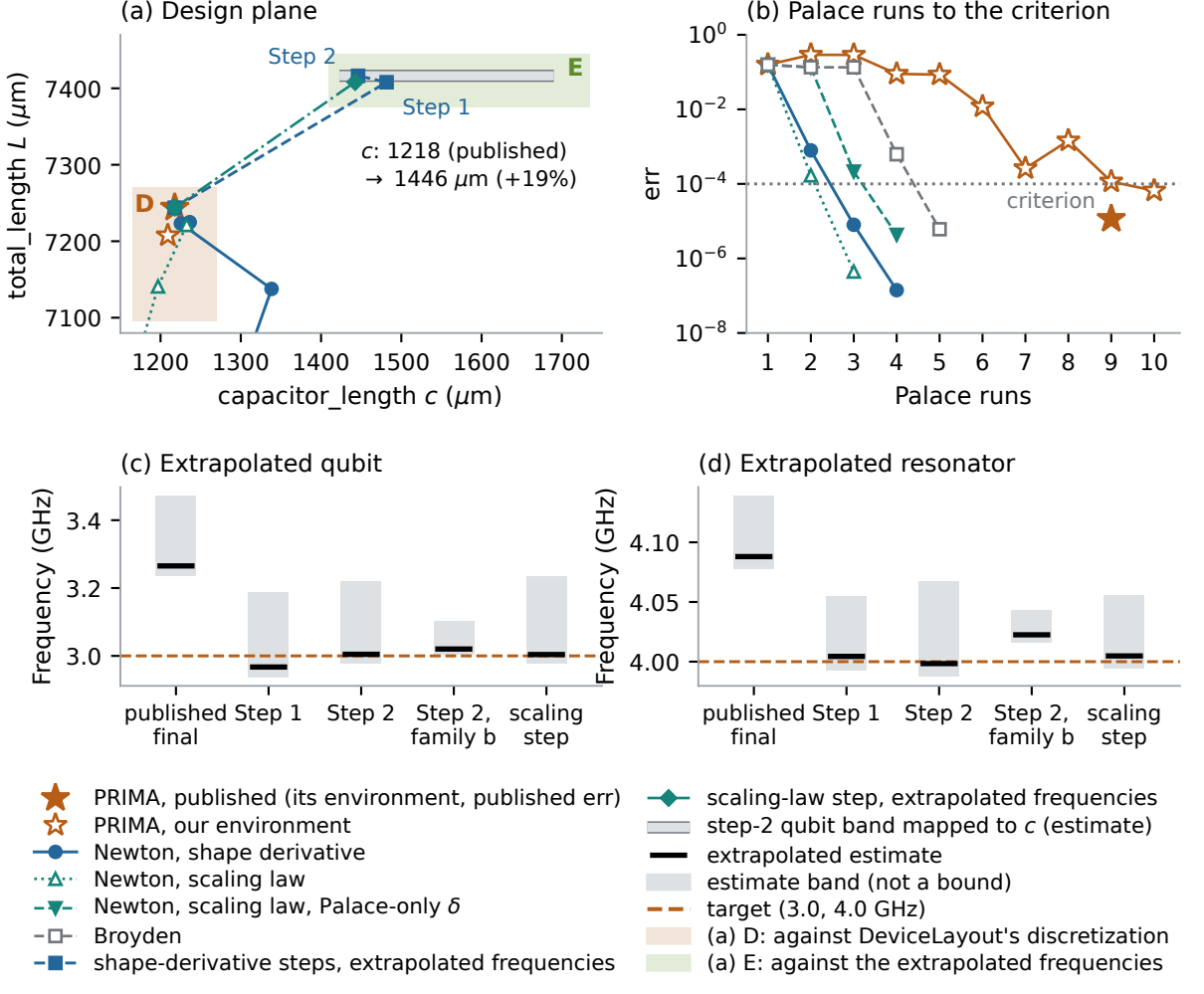
(a) Design plane
Step 2
E
Step 1
c: 1218 (published)
→ 1446 μm (+19%)
D
total_length L (μm)
7400
7300
7200
7100
1200
1300
1400
1500
1600
1700
capacitor_length c (μm)
(b) Palace runs to the criterion
err
10^0
10^-2
10^-4
10^-6
10^-8
criterion
1
2
3
4
5
6
7
8
9
10
Palace runs
(c) Extrapolated qubit
Frequency (GHz)
3.4
3.2
3.0
published final
Step 1
Step 2
Step 2, family b
scaling step
(d) Extrapolated resonator
Frequency (GHz)
4.10
4.05
4.00
published final
Step 1
Step 2
Step 2, family b
scaling step
PRIMA, published (its environment, published err)
PRIMA, our environment
Newton, shape derivative
Newton, scaling law
Newton, scaling law, Palace-only δ
Broyden
shape-derivative steps, extrapolated frequencies
scaling-law step, extrapolated frequencies
step-2 qubit band mapped to c (estimate)
extrapolated estimate
estimate band (not a bound)
target (3.0, 4.0 GHz)
(a) D: against DeviceLayout's discretization
(a) E: against the extrapolated frequencies